

Combining Revealed and Stated Preference Methods for Valuing Environmental Amenities*

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Received October 26, 1992; revised February 9, 1993

A stated preference model and a revealed preference model for recreational site choice are examined and compared. Both models are based on random utility theory and the data are obtained from the same individuals. The stated preference model is based on the respondent's choice from hypothetical choice sets. Attributes in the stated preference model are based on the ranges of the actual levels of attributes in the revealed preference choice set and are presented to respondents using a fractional factorial statistical design. The results show that while independently estimated models appear to reflect different underlying preferences, joint estimation of the model parameters, including estimation of the relative scale parameter, provides evidence that the underlying preferences are in fact similar. Furthermore, combining the revealed and stated preference information yields other benefits in estimation. © 1994 Academic Press, Inc.

INTRODUCTION

Methods for valuing environmental amenities have traditionally been categorized as indirect and direct. Indirect methods, like the travel cost model, use actual choices made by consumers to develop models of choice. These constitute revealed preferences over goods, both market and non-market. Direct methods ask consumers what they would be willing to pay or accept for a change in an environmental amenity. Direct methods are examples of stated preference techniques in that individuals do not actually make any behavioral changes, they only state that they would behave in this fashion.

Both methods have advantages and drawbacks. Direct methods are commonly criticized because of the hypothetical nature of the questions and the fact that actual behavior is not observed (Cummings *et al.* [10], Mitchell and Carson [29]).

*The authors thank Peter Boxall, John Thompson, Ted Graham-Tomasi, two anonymous reviewers, and the participants of seminars at the University of Guelph and Michigan State University for their comments on this paper. Funding support from Alberta Environmental Protection is gratefully acknowledged. Authors are listed alphabetically, senior author is not assigned.

However, direct methods currently provide the only viable alternative for measuring non-use values and they are commonly used to elicit values in cases in which the environmental quality change involves a large number of attribute changes. Indirect methods avoid the criticism of being based on hypothetical behavior, but the models of behavior developed constitute a maintained hypothesis about the structure of preferences which may or may not be testable. Indirect methods may also suffer on the grounds that the new situation (after the environmental quality change) may be outside the current set of experiences (or outside the data range) and thus simulation of the new situation would involve extrapolation outside the range used to estimate the model. Finally, indirect methods may suffer from colinearity among attributes. Colinearity precludes the isolation of factors affecting choice. This isolation is often required in economic welfare analysis. For example, water quality attributes (BOD, turbidity, etc) may be correlated but the economic valuation may only be interested in valuing an improvement in one of the attributes. The separation of these attributes is necessary for an accurate representation of the benefits and for policy analysis.

In the non-market valuation literature, indirect methods (primarily travel cost models) and direct methods (primarily contingent valuation) have been used to value environmental quality changes. While contingent valuation is the most common stated preference approach found in the literature, some researchers have employed contingent ranking or contingent behavior (e.g., Lareau and Rae [16], Rae [31], or Smith and Desvousges [33]). The stated preference application we present here can be considered a type of contingent behavior analysis. We ask individuals to respond to a set of questions which require them to choose one alternative from three options. In each question, the individual is faced with choices of a running water recreation site, a standing water recreation site, and the choice to participate in some other non-water-based recreation activity or stay at home. The running and standing water alternatives are described by a set of characteristics of the recreational sites. These characteristics include such items as the distance to the site, the water quality, and the fishing catch rate.¹ The choice made by the individual indicates a preference for the attributes of one alternative over the others.

Each individual completes a series of these choice questions. In each question they are presented with running and standing water sites described by different levels of the attributes of the sites. The levels of the attributes are chosen based on a statistical design which results in orthogonal attribute data. The use of factorial statistical designs has been employed in the non-market valuation literature by Goodman [13] in a hedonic price example and Mackenzie [23] in a recreational hunting example. In these studies, however, the statistical design was used to produce descriptions of "goods" which were then rated or ranked by the respondents. This type of approach, common in the conjoint analysis literature, has been criticized by Freeman [12]. In our approach, we ask the individuals to choose one of the presented alternatives, making the process consistent with random utility theory and parallel to site choice models which employ discrete choice random utility theory (e.g., Smith [32], Bockstael *et al.* [6]).

¹The complete set of attributes and further details regarding the choices of the attributes and levels are described below.

In this paper we use the stated preference approach described above to characterize recreationist choice of sites. The attributes used to describe the sites in the stated preference model are chosen based on the actual ranges of attributes of the sites in the study region. These ranges of actual values are used to develop categories or levels of the attributes. For example, distances traveled by recreationists living in this region range from about 25 to about 150 km. A set of four levels of distance (25, 50, 100 and 150 km) were used in the stated preference experiment to describe the attribute distance. We also elicit information on actual choice from these recreationists. This information is used to develop a discrete choice model of actual site choice, based on the same type of factors as the stated preference model. Thus, the two techniques are using similar factors to explain choice: revealed choice and stated choice. However, the stated preference method employs a statistical design which eliminates colinearity among the attributes. The attribute levels presented to the respondent are chosen using principles from the design of fractional factorial experiments frequently used in marketing research applications of choice-based conjoint analysis, which we explain in more detail in the section, Research Approach (Louviere and Hensher [17, 18], Louviere and Woodworth [19], Louviere [20, 21, 22], Batsell and Louviere [2]). While this model has been applied to the analysis of demand for market goods (e.g., Bunch *et al.* [7], Louviere and Hensher [18], Louviere and Woodworth [19], Louviere [22]), to our knowledge it has not been widely used in the non-market valuation field.

Both the revealed preference and the stated preference models are discrete choice models. In the revealed preference case the individuals choose one recreation site from all sites available to them. The model explains the choice of site as a function of travel distance and quality attributes of the sites. The stated preference model explains the choice of one alternative over the other as a function of the attributes which include travel distance and the same quality attributes. Both models reflect the same trade-offs and both can be considered applications of random utility theory.

Since the two approaches reflect the same process of choosing recreation sites based on attributes, it is possible to combine the analyses. The advantages of jointly estimating the model are the following. First, the stated preference information reduces colinearity that may be present in the revealed preference data set. Therefore, attribute effects that were previously unidentified or weakly identified due to colinearity can now be more clearly identified. Second, in cases where the actual data do not encompass the range of the proposed changes, stated preference experiments can be used to include the range of the proposed changes within the data series.

The stated and revealed preference models are estimated using multinomial logit techniques. It is widely recognized that when operating in a random utility context, the scale parameter in the error term is arbitrarily assumed to be unity. This assumption has no effect on the utility levels within each of the separate revealed preference and stated preference models. However, the scale parameters could be quite different between the two models, reflecting the respective levels of variation in the two different data sets (Ben-Akiva and Lerman [5, pp. 104–106]). In the analysis which follows we show that there is a significant correlation in the underlying preferences as described by the predicted proportions of visits to each site. The difference between the two models appears to lie in the scale factor. Therefore, we estimate the joint model and simultaneously estimate the relative

scale factor for the stated preference data which results in the highest likelihood value. With two separate data series it is possible to estimate the scale factor for one of the series, jointly with the other parameters, as the value resulting in the highest likelihood (Ben-Akiva and Morikawa [4], Swait and Louviere [34]).

Each technique produces results that are based on random utility theory, hence, either can be used for welfare analysis of environmental quality changes. Similarly, the joint model can be used for welfare calculations. In the application presented below, we applied stated and revealed preference models to the evaluation of alternative flow scenarios for the Highwood and Little Bow rivers in southwestern Alberta. These flow scenarios were developed to better satisfy instream flow needs (IFN) for recreation, fisheries, wildlife, and stream ecology, as required by the Water Management Policy for the South Saskatchewan River Basin. We employ the welfare economic propositions derived from the random utility model to estimate the economic value of the improvements in environmental quality associated with the flow scenarios. Because the stated preference, revealed preference and joint models are all based on random utility models, we can compare the results of the three approaches.²

The rest of the paper proceeds as follows. The next section contains a description of the stated preference approach, followed by a brief review of the revealed preference approach. The data and empirical results are then presented.

STATED PREFERENCE MODEL

Stated preference models are common in the marketing and applied decision research literature (see [20]). These techniques are often referred to as "experimental or stated choice analysis" or "conjoint techniques." A case for their use in market research and economic analysis has been presented by McFadden [25]. Stated preference approaches involve asking respondents to rank or judge attributes or products or asking respondents to choose from hypothetical choice sets [8, 20, 21, 22]. The approach we use is to develop hypothetical choice sets in which the choices are described by bundles of attribute values associated with water-based recreation.

In order to design the choice sets, a set of attributes affecting the choice of recreational sites was developed to reflect actual characteristics of water-based recreational resources in the study area. These 13 attributes and the levels chosen for our analysis are listed in Table I. In our study area, recreational fishing is a significant form of recreation and thus the attribute list contains several variables relating to fishing activity and success. The list also includes travel distance and other features relating to the scenic value of the sites. Also, some attributes are alternative specific, such as the water feature. This attribute depends on a higher level of attribute, running or standing water. Note that the attribute levels varied in the choice sets are discrete even though in some cases they reflect underlying continuous variables. Discrete attribute levels are a consequence of the statistical design process used to create the choice sets.

²For illustration, we also provide some examples of welfare changes brought about by changes in single quality attributes, i.e., a change in travel cost or a change in fish catch rate.

TABLE I
List of Attributes Used in the Stated Preference Experiment

Attribute	Description
Terrain	Flat prairie Rolling prairie Foothills Mountain
Fish size	Large Small
Fish catch rate	1 fish per 4 hours 1 fish per 80 minutes 1 fish per 45 minutes 1 fish per 35 minutes
Water quality	Good Bad
Facilities	None Day-use only Limited facility campsite Fully serviced campsite
Swimming	Yes No
Beach	Yes No
Distance	25 km 50 km 100 km 150 km
Standing water	
Water feature	Natural lake Reservoir
Fish species	Pike and perch Pickerel, pike, and perch
Boating	None Small crafts Power boats (limited) No restrictions
Running water	
Water feature	River Stream
Fish species	Mountain whitefish Rainbow trout and mountain whitefish Rainbow trout, mountain whitefish, and brown trout Cutthroat trout, mountain whitefish, and bull trout

Research Approach

The set of attributes and levels displayed in Table I can be viewed as setting the space to be spanned in the choice experiment. If we treat each attribute as discrete, as in Table I, there are $2^6 \times 4^5$ possible standing water alternatives, and an additional $2^5 \times 4^5$ possible running water alternatives. The problem of choice set construction, therefore, can be viewed as sampling from the universe of possible pairs of standing and running water alternatives. Methods of design construction for such problems have been considered by Louviere and Hensher [18], Louviere and Woodworth [19], Louviere [20, 21], Batsell and Louviere [2], and Bunch *et al.* [8] under the assumption that the choice process can be described by McFadden's [27] "Mother" logit model or its special cases, such as conditional logit [26]. Design strategies described in these sources are consistent with various subset forms of the more general Mother logit form and produce consistent estimates of the parameters of these models. In fact, McFadden [25] has referred to these design strategies as "powerful" methods for estimating and testing violations of the IIA property of simple, conditional logit models.

In the present case, we wished to ensure that we could test for violations of the IIA property of conditional logit. To do this, we designed the survey such that respondents always had to choose among (1) a standing water option, (2) a running water option, or (3) staying at home and not participating in water-based recreational activity. Testing IIA requires at least three alternatives, which is satisfied in this case by having the "stay at home" option. As discussed by Louviere [20, 21] and Louviere and Woodworth [19], the minimum requirement for testing IIA is that the attributes of the choice alternatives be orthogonal within and between alternatives. One way to satisfy this requirement is to treat the attributes of standing and running water as a collective factorial, i.e., a $(2^6 \times 4^5) \times (2^5 \times 4^5)$, and select an orthogonal main effects design to vary all attribute levels simultaneously. A main effects design is one in which only strictly additive variance components can be estimated, assuming that all interactions equal zero.

In practice, even the smallest orthogonal main effects design will permit one to estimate all the strictly additive (i.e., main effects) parameters of the Mother Logit model. These parameters include generic or alternative-specific effects for all attributes, together with alternative-specific effects of the attributes of one alternative on other alternatives (e.g., running water distance on choice of standing water). The IIA property of conditional logit requires that such "cross-effects" equal zero in principle or are non-significant in practice. Hence, the ability to incorporate such terms in the model specification permits a powerful test of IIA (McFadden [25], McFadden *et al.* [28]).

This design strategy produces consistent but less than optimally efficient estimates of the logit parameters. Statistically efficient designs for choice experiments based on the Mother Logit model or its nested relatives like conditional or nested logit remain elusive because optimizing efficiency requires knowledge of the true parameter values in advance. Recent work by Bunch *et al.* [8] suggests that designs like the one employed in this study are relatively but not optimally efficient. The total number of parameters that could be estimated in the present case are [standing water main effects, including all linear, quadratic, and cubic components (6 + 15)] + [running water main effects, including all linear, quadratic, and cubic components (5 + 15)] + [cross-effects of standing water attributes on choice of

running water (6 + 15)] + [cross-effects of running water attributes on choice of standing water (5 + 15)] + [alternative specific constants for standing and running water, respectively (2)] = 84 possible parameters. In practice, many of the cubic components should not be significant and hence are not estimated, reducing the estimation problem.

We present only the results of estimating the main effects models, leaving aside the issue of cross-effects tests of IIA. The final design, therefore, consisted of 64 choice sets chosen from the $2^{11} \times 4^{10}$ full factorial of possible attribute level combinations using the main effects design strategy explained earlier. This design is derived by using main effects and interaction columns in the 4^3 to define the attributes of standing and running water alternatives in such a way that all parameters of interest can be estimated if we assume that all omitted interaction effects equal zero. In essence, the design tells us exactly how to define the 64 pairs

Recreational Opportunities

Suppose last August that you could have chosen only from the recreational opportunities described below.

	A. Standing	B. Running	C. Non-Water
Water Feature	Natural Lake	Stream	
Terrain	Mountain	Foothills	
Driving Distance	50 km	50 km	
Fishing:			Any other non-water related recreational activity or stay at home.
Types of fish available	Pickeral, Pike & Perch	Rainbow Trout & Mtn. Whitefish	
Fish size	Large	Large	
Typical fishing success	1 fish every 35 min.	1 fish every 4 hours	
Camping Facilities	Designated Campsite	Designated Campsite	
Water Quality	Good	Good	
Boating	None	None	
Swimming	Yes	No	
Beach	No	Yes	
Day Use or Entry Fee to Maintain Facilities	None	\$6	

1. If these recreational opportunities had been available to you last August, would you have seriously considered visiting

Opportunity ☐ Yes ☐ No
 Opportunity ☐ Yes ☐ No

2. Had the above opportunities been available last August, which one would you have most likely chosen? (Check one and only one box)

☐
A

☐
B

☐
C

FIG. 1. Example of stated preference question.

of standing and running water alternatives such that the model parameters of interest can be estimated. Recalling that there are 84 possible parameters of estimation interest, the 64 choice sets provide 3×64 choice observations (times the number of respondents observed in each set), ensuring ample degrees of freedom to estimate all parameters should that prove necessary.

It might also be noted that logit models are "difference-in-utility" models. That is, logit model parameters are defined on design matrix columns that represent differences in attribute levels for continuous attributes or contrasts in attribute levels for qualitative attributes. The fractional factorial design strategy described above orthogonalizes the absolute attribute levels, not differences. However, as discussed by Louviere [20, 21] and Louviere and Woodworth [19], the addition of a constant reference alternative to each choice set—in the present case "stay at home"—results in the differences being calculated with respect to a constant. Hence, the orthogonality of the original design is preserved due to this "trick." Figure 1 displays an example choice set and task instructions to illustrate how the design was implemented into a survey format.³

It is unrealistic to assume that individuals can or will respond to all 64 choice sets in an interview setting, especially a mail survey setting of the type we employed in this research. Consequently, we blocked the experiment into four sets of 16 choice sets by using an additional four-level column as a factor in the design. Blocking in this manner ensures that each block of choice sets is approximately statistically equivalent. The 16 choice sets in each block were placed into a survey format, which was described as a "recreation game." As part of the survey materials, each respondent received a set of instructions for completing the survey and the choice task, a glossary describing the meaning of each of the attributes and levels in plain English, and a telephone contact number for additional assistance. The respondent's task was to choose between either a standing or running water alternative described in each choice set or to decide to choose not to engage in water-based recreation for their next one-day or overnight outing.

Data

The data for both the revealed preference analysis and the stated preference analysis were obtained from a multiphase survey sponsored by Alberta Environment. Random digit dialing techniques were used to contact individuals within the study region (an area in and around the city of Calgary, Alberta). A total of 4497 numbers were called. Subtracting disconnected numbers and no-answer calls, the total number of individuals contacted was 2383. Of those contacted, 1232 (51.7%) completed the telephone portion of the survey. In this telephone portion, individuals were asked if they participated in water-based recreation, where they participated in such activities, what type of recreation they engaged in, and a variety of socioeconomic characteristics.⁴ Each person who responded that they participated

³Note that the stated preference question asks which alternative, including the no choice option, the respondent "would have most likely chosen." One could also ask "which site would you choose" to coincide more closely with the contingent behavior approach. We suspect that there will be little difference between these two forms of questions.

⁴Socioeconomic characteristics elicited included age, gender of the respondent, education level, and family structure. Due to sensitivities within the funding agency, individual or household income information was not collected.

in water-based recreation was asked if they would be willing to complete a mailed-out questionnaire which contained the stated preference experiment. Of the 1232 contacts, 785 (63.7%) agreed to receive the mail-out portion of the experiment. After a period of 3 weeks, 413 accurately completed mail-out portions were returned (45%).⁵ See Alberta Environment [1] for details on the survey.⁶

Individuals were asked about the number of water-based recreation trips they took in the previous year. They were also asked to discuss specific details of the trips they took in August of 1991 (the survey was administered in the fall of 1991). Seventy-four percent (911) of the 1232 respondents participated in water-based recreation. The respondents took an average of 11.85 trips per year (or slightly under 1 trip per month) in total. As described above, the study focused on recreation trips in August 1991 and on 20 recreation sites within the study region (a set of sites which typically capture the majority of the water-based recreation outings in the region). During the month of August 1991, these respondents took a total of 730 trips to recreation sites to the 20 sites. These 730 trips are the basis for the revealed preference analysis, described below.

The data used for the stated preference analysis were obtained from the mail survey component. A recreation game containing the 16 sets of choices described above was mailed to those indicating that they would be willing to participate in this portion of the survey. Four hundred thirteen respondents provided complete answers to the choice experiment.

Analysis

The response to the choice between three constructed recreation choice alternatives can be modeled in a random utility framework. The overall utility can be expressed as the sum of a systematic component, which is expressed as a function of the attributes presented, and a random component. More formally, this can be expressed as

$$U_{in} = v_{in} + e_{in}, \quad (1)$$

where U_{in} is person n 's utility of choosing site (alternative) i , v_{in} is the systematic component of utility, and e_{in} is a random element. Site (alternative) i is chosen over site j if $U_{in} > U_{jn}$. The probability of individual n choosing site i is

$$\pi_n(i) = \Pr\{v_{in} + e_{in} \geq v_{jn} + e_{jn}; \forall j \in C_n\}, \quad (2)$$

⁵The mail-out portion of the survey was handled by an agency independent of the agency performing the telephone survey. Due to an unfortunate set of circumstances, there was no link between the respondents to the telephone survey and the mail survey. All that is known is that the respondents to the mail survey are a subset of the respondents to the telephone survey. Thus, socioeconomic characteristics collected in the telephone survey could not be used in the development of the stated preference model. While this is an unfortunate aspect of the survey approach used here, it does not affect the methods employed. However, it does suggest that the results presented here should be viewed as an example of the approach rather than the ultimate estimation of the welfare changes involved.

⁶Some of the descriptive statistics associated with the telephone survey follow: 50.6% of the respondents contacted were male; the average respondent had a high school education (46% of respondents); the average household size was 2.7 individuals; and the family structure of the majority of the households contacted was couples with children (48%).

where C_n is the choice set for individual n . v_{in} is a conditional indirect utility function and we assume it has a linear form,

$$v_{in} = \beta_1 + \beta_2 x_{in2} + \beta_3 x_{in3} + \cdots + \beta_k x_{ink} + \alpha(Y - P_i), \quad (3)$$

where x_{ink} are attributes of the site (alternative), Y is income, and P_i is the price (or travel cost) of accessing the site. Assuming that the error terms are Gumbel distributed with scale parameter μ , the probability of choosing site i is

$$\pi_n(i) = \frac{\exp^{\mu V_{in}}}{\sum_{j \in C_n} \exp^{\mu V_{jn}}}. \quad (4)$$

Note that the scale factor μ remains in this expression and is typically assumed to equal 1. However, since we have two separate data sets, it is possible to jointly estimate the relative scale factor (or ratio of the two scale parameters) for one set along with the model parameters for the joint data. A more detailed discussion of this procedure is presented below.

Formulated in such a manner, the stated preference model is described exactly like a revealed preference travel cost model. Welfare measures from that model can be developed using the theory applied to revealed preference discrete choice models (Hanemann [14]). A measure commonly used to examine the impact of a quality change is

$$CV = -\frac{1}{\alpha} \left\{ \ln \left(\sum_{i \in C_n} \exp(v_{n0}) \right) - \ln \left(\sum_{i \in C_n} \exp(v_{n1}) \right) \right\}, \quad (5)$$

where α is interpreted as the marginal utility of income, v_{n0} is the initial state (or quality level), and v_{n1} is the level of utility in the subsequent state.⁷ In the empirical analysis presented below, the parameters of the discrete choice model are estimated using the stated preference data. These parameters are then applied to determine the choice probabilities for each site in the actual area under investigation. The attribute levels associated with the various flow scenarios are used to generate measures of welfare.

In the stated preference model attributes with multiple levels are coded using effects codes.⁸ The base level is assigned -1 for all columns representing the remaining levels. Each column contains a 1 for the level represented by the column and a -1 for the base. For example, in the fish species case, mountain whitefish (base) are assigned a value -1 for all three columns of fish species. The first column represents rainbow trout and mountain whitefish (coded 1) versus mountain whitefish alone (coded -1). The interpretation of these parameters is that the base level takes the utility level of the negative of the sum of the estimated coefficients and each other level takes the utility associated with the coefficient.

The convention in stated preference models is to use "effects codes" rather than 1,0 dummies. Effects codes are common in applied statistics in the analysis of designed experiments because (a) 1,0 dummies confound the alternative-specific

⁷This welfare measure examines the impact of a change for a single choice occasion. In our models we assume that each trip is a choice occasion and we examine the impact of changes on trips from various residence areas (Table V).

⁸Effects coding is also used in the joint model described below.

constant with the effects of interest; whereas, effects codes orthogonalize the attribute effects to the constant, (b) effects codes simply contrast the parameter estimates with one of the levels; whereas 1,0 dummies contrast the estimates with the constant, and (c) interactions defined from effects coded columns are orthogonal to their respective main effects and other estimable interaction effects; whereas 1,0 coded dummies are not. Thus, either coding scheme should produce the same estimates up to a positive linear transformation; however, effects codes have more desirable estimation properties, particularly when one is dealing with small design spaces in which the number of observations relative to the number of parameters is small.

Results

The results of the logit analysis of the stated preference data are presented in Table II. The model has a very good fit (McFadden's $\rho^2 = 0.76$), and the parameters have signs consistent with our expectations. All parameters except distance are estimated as interactions with the water feature (standing or running water). Larger fish, increased catch rates, good water quality, and the availability of swimming and beaches are factors which positively affect utility. Mountains and foothills are preferred to flat prairie and rolling prairie. More fish species are preferred to fewer with the package of rainbow trout, mountain whitefish, and brown trout providing the highest utility. Fully serviced campsites are preferred on both standing and running water sites. Distance is a negative factor as expected. Note that there are differences between the running and standing water coefficients. An increase in fish catch or size, for example, would produce a larger increase in utility at running water sites than at standing water sites.

The parameters from the stated preference model were estimated from the respondent's choice from a set of three hypothetical alternatives. However, the attributes can be associated with actual recreation sites and the model can be used to determine the probabilities of choosing each site. A choice set, which reflects the set of sites from which the consumer can choose, must be defined. For this study, the choice set was defined as 24 recreational sites in the region of the project. These 24 sites were chosen on the basis of their role as substitutes for each other and their location both within the study region and approximately 2 hours driving time from the population centers in the region.

To evaluate the current conditions, attributes for the 24 sites were combined with the parameters from the model to produce a distribution of trips predicted by the stated preference model. This set of sites represents the major recreational reservoirs and rivers in the region. This combination of actual attributes and stated preference model parameters was also used to create the utilities necessary for the welfare calculation. To evaluate the effects of changing river flows to better satisfy instream flow needs, the attributes were changed to reflect the hydrological and biological conditions associated with the new flow regimes, based on information provided by various environmental consultants. In general, the modified flow regimes are expected to affect fish catch rates and species composition at sites along the Highwood and Little Bow rivers. Also included in the analysis is the construction of an on-stream reservoir that would include opportunities for fishing, boating, and day-use recreation and improve an existing lake (Clear Lake) which is currently not fishable.

TABLE II
Coefficients of the Stated Preference Model

Attribute	Description	Standing water ^a	Running water ^a
Terrain ^b	Flat prairie (= -1) versus	-0.367	-0.415
	Rolling prairie (= 1)	-0.071 (0.046)	-0.100 (0.042)
	Foothills (= 1)	0.257 (0.045)	0.125 (0.042)
	Mountain (= 1)	0.181 (0.045)	0.390 (0.042)
Fish size	Large (= 1) vs small (= -1)	0.058 (0.026)	0.090 (0.025)
Fish catch	Fish per unit time	0.062 (0.028)	0.105 (0.026)
Fish species ^b	Mountain whitefish (= -1) versus	n/a	-0.275
	Rainbow trout and mountain whitefish (= 1)	n/a	0.064 (0.043)
	Rainbow trout, mountain whitefish, and brown trout (= 1)	n/a	0.107 (0.041)
	Cutthroat trout, mountain whitefish, and bull trout (= 1)	n/a	0.103 (0.042)
Water quality	Good (= 1) vs Bad (= -1)	0.394 (0.027)	0.321 (0.025)
Facilities ^b	None (= -1) versus	-0.353	-0.277
	Day-use only (= 1)	-0.200 (0.046)	-0.109 (0.043)
	Limited facility campsite (= 1)	0.305 (0.045)	0.162 (0.042)
	Fully serviced campsite (= 1)	0.248 (0.045)	0.225 (0.042)
Swimming	Yes (= 1) vs no (= -1)	0.274 (0.026)	0.158 (0.025)
Beach	Yes (= 1) vs no (= -1)	0.198 (0.026)	0.123 (0.024)
Distance	Kilometers	-0.007 (0.0004)	-0.007 (0.0004)

^aWater feature-specific (running versus standing water) coefficients are estimated for most attributes except distance. n/a indicates not applicable.

^bAttributes with multiple levels are coded using effects codes. The base level is assigned -1 for all columns representing the remaining levels. Each column contains a 1 for the level represented by the column and a -1 for the base. The interpretation of these parameters is that the base level takes the utility level of the negative of the sum of the estimated coefficients and each other level takes the utility associated with the coefficient.

The welfare measures were calculated from the expression in Eq. (5). Per trip welfare measures were calculated using the absolute value of the coefficient of travel cost as the marginal utility of income. The resulting welfare measures are presented for each residence zone. Unfortunately, no socioeconomic data on respondents except residence zone were available for the analysis. Therefore the welfare results were identical for respondents from a residence zone but varied across residence zones. The welfare measures are presented in Table V. Welfare measures are presented without including time values (based only on a travel cost of \$0.27 per kilometer). These measures are underestimates of the true value since the value of time is not included.⁹ The per trip welfare measures vary by residence zone from a maximum of \$8.06 for residents of zone 21 (travel cost only) to a minimum of \$4.33 (zone 10 residents). These variations are the result of distance to the sites.¹⁰

REVEALED PREFERENCE ANALYSIS

Information on the actual choices of recreation sites was collected from the same sample of individuals who provided responses to the stated preference survey. Therefore, a comparison between the stated preferences and the revealed preferences for the same set of individuals is possible.

Data

As described above, the individual trip response variables were obtained from the recreation telephone survey conducted for Alberta Environment. Within the telephone survey individuals were asked to report the destination of the recreation trips they took during the month of August 1991. These trip destinations were coded to correspond with the sites in the study area. The sample produced information on 730 trips which were included in the statistical model.

Attribute levels for the various sites were the actual levels (for each site) of the attributes used in the stated preference model. This information was obtained from various consultant reports as well as expert judgement by staff at Alberta Environment and Alberta Forestry, Lands and Wildlife. Travel distances were calculated from each residence zone to each recreation site.

A multinomial logit discrete choice model of site choice was specified and estimated. Travel cost (distance) and other site attributes were used to explain site choices. The quality factors could not all be entered into the statistical analysis. Some factors reflected similar information for the sites and therefore could not be

⁹As described above, individual or household income was not available. Therefore we could use only zonal average income estimates or estimates based on average wage rates to measure values of time. For this particular application we have not measured these values but we recognize that such a calculation should be performed to develop a more accurate estimate of the welfare effect.

¹⁰As an illustration of the approach, we also computed welfare measures for a 10% increase in travel costs to all sites and a 10% increase in catch rate at all sites. The change in travel costs results in an average (across sites) loss of \$3.48 per trip (revealed preference), \$4.68 per trip (stated preference), and \$3.60 per trip (joint model). A 10% increase in the fishing catch rate results in an average benefit of \$1.74 per trip (revealed preference), \$0.11 per trip (stated preference), and \$0.43 per trip (joint model).

TABLE III
Coefficients of the Revealed Preference Model

Attribute	Description	Standing water ^a	Running water ^a
Distance	Kilometers	-0.0282 (0.0013)	-0.0282 (0.0013)
Catch	Fish per unit time	2.0338 (0.2371)	2.0338 (0.2371)
Swimming	Yes (= 0) vs no (= 1)	2.7477 (0.2896)	0.9148 (0.2505)
Beach	Yes (= 0) vs no (= 1)	0.9918 (0.3019)	-1.955 (0.3687)
Water quality	Poor (= 1) vs good (= 0)	-0.8197 (0.4941)	-3.129 (0.3749)
Boating	Unrestricted (= 1) versus none	6.6620 (1.024)	1.7335 (0.2886)
	Power boats on lake sites (= 1) versus none	7.2496 (1.062)	n/a n/a
Water feature	Running water (= 1) versus standing water	n/a	7.6178 (1.1290)
Site constant 2	Site-specific shift constant	2.8340 (0.3993)	2.8340 (0.3993)
Site constant 13	Site-specific shift constant	2.9120 (0.2920)	2.9120 (0.2920)
Site constant 14	Site-specific shift constant	2.6021 (0.2953)	2.6021 (0.2953)
Site constant 17	Site-specific shift constant	0.6878 (0.1845)	0.6878 (0.1845)

^aWater feature-specific (running versus standing water) coefficients are estimated for most attributes except distance, catch per hour, and the site-specific constants. n/a indicates not applicable.

separated in the statistical analysis. The resulting model contains a subset of the parameters estimated in the stated preference model.

Results

The model was estimated using a variety of combinations of attributes and the most complete model (presented in Table III) was used to analyze the welfare impacts of the changes introduced by the project.¹¹ As in the stated preference

¹¹While all factors can be included in the stated preference model (and the joint model), colinearity prevents the inclusion of all factors in the revealed preference model. In some cases, attributes were perfectly colinear and could not be included, while in other cases, severe, but not perfect, colinearity resulted in fragile coefficient estimates. Also, some attributes did not have enough variation between sites to provide significant coefficient estimates. The model presented here is a specification that is relatively robust and includes the elements required for the welfare analysis (e.g., analysis of the post-project conditions).

model, most parameters were estimated with separate coefficients for the attribute on running water and on standing water. The revealed preference model has a McFadden's $\rho^2 = 0.28$, which is considerably lower than the goodness of fit measure for the stated preference model. This probably results from the fact that the stated preference model was estimated in aggregated form (using frequencies) and the fact that there are many more elements which influence site choice in the real world, while in the stated preference model the choices are somewhat simplified and focused.

The statistical model indicates that distance, water quality, fishing success, presence of swimming areas, the availability of boating, and the presence of beaches are significant factors influencing site choice. The sign on distance is negative, indicating that sites are less attractive if they are further away from the respondent's home. The signs on fishing success and boating variables are positive, indicating that the availability of these attributes (or higher levels of success) positively influences choice. Water quality coefficients are negative because the variable is 1 for poor sites and 0 for good sites. The site-specific shift constants are included to account for factors not collected in the quality attribute database that influence choice of these particular sites. These four site-specific constants are chosen because they represent some of the more popular sites in the choice set. Tests of the statistical properties of this model show that the model's parameters are highly significant in explaining choice.

The model presented above is used to estimate the economic impact associated with the proposed changes in river flows. The economic welfare measures are calculated using the expression in Eq. (5). Also, as mentioned above, travel distance is converted to travel cost by applying a factor of \$0.27 per mile.

The resulting welfare measures, for each residence zone, range from \$0.46 per trip (residents of zone 8) to \$3.99 per trip (zone 20). Note that while the stated preference welfare measures are higher than the revealed preference measures for each residence zone, the differences are not constant. The revealed preference measure for zone 8 is 8 times smaller than the corresponding stated preference measure. On the other hand, the stated preference measure for zone 20 is only 2 times the revealed preference measure. The models represent a different quantification of the trade-offs between site attributes, or scaling differences due to different magnitudes of the respective random components.

JOINT REVEALED – STATED PREFERENCE ANALYSIS

The revealed preference model suffers from the fact that not all quality attributes can be included in the model because the actual data are colinear. However, the stated preference model is designed to eliminate this effect. Joint estimation of the revealed and stated preference data can remove the effect of colinearity from the revealed preference data. The joint estimation procedure can be interpreted as a type of mixed estimation (Belsley *et al.* [3]; Ben-Akiva and Morikawa [4]) since the stated preference data add external information about the data and/or parameters of the model.¹² Joint estimation adds information through

¹² In this case we interpret the procedure as the use of stated preference data to add information to the revealed preference estimation. However, the reverse is also possible.

the addition of additional data points. However, joint estimation also makes it possible to estimate the relative scale factor for the two data series.

The joint estimation is accomplished in the following steps: (1) Separate estimation of revealed and stated models and (2) concatenation of both data sets, and rescaling of stated data relative to revealed by conducting a grid search in the manner suggested by Swait and Louviere [34]. That is, (a) multiply the columns of the stated design matrix by a constant, beginning at one end of the search range, (b) estimate the model on the joint data sets and obtain the log likelihood for the joint model, (c) repeat by incrementing the constant, (d) stop at that constant value that maximizes the likelihood value, and (e) compare the joint likelihood to the sum of the separate likelihoods for the stated and revealed models. Accept the null that the rescaled parameters are the same if the joint and summed separate likelihoods do not differ statistically.

This procedure is possible because the probabilities of choosing i in the conditional logit model for the revealed and stated data are:

$$\begin{array}{ll} \text{Revealed:} & \pi(i) = \exp(\mu_r V_i) / \sum_j \exp(\mu_r V_j) \\ \text{Stated:} & \pi(i) = \exp(\mu_s V_i) / \sum_j \exp(\mu_s V_j), \end{array}$$

where μ_r is the scale parameter for the revealed preference data set and μ_s is the scale parameter for the stated preference data set.

For any one data set the μ 's cannot be identified and are set to one for convenience. However, in any pair of data sets, their ratio, i.e., μ_r/μ_s , can be identified, which is what is estimated in the joint estimation. Unfortunately, the Swait and Louviere [34] procedure does not provide a test statistic for the estimated ratio; although, as they indicate, an approximate confidence interval for the estimate can be derived. Only if the null hypothesis of no significant parameter differences can be retained after rescaling can the ratio be interpreted as a ratio of random components. Otherwise, it is simply an optimal rescaling constant that maximizes the fit of the stated and revealed parameters given the conditional logit model.

The colinearity is "reduced," not eliminated, because the design matrix for the stated data is orthogonal. Hence, even if the revealed data are highly colinear, this colinearity must be reduced by the addition of the orthogonal columns. Similarly, in the revealed data some of the destination-specific constants were perfectly colinear with certain attribute values. The addition of the stated data now reduces that colinearity and makes it possible to identify values for these constants.

As described above, if one performs joint estimation it is possible to estimate the ratio of the two scale parameters in the model. The process jointly determines the coefficients applied to the attributes and the relative scale factor for one part of the data that homogenizes the two variance components. In this case we chose to rescale the stated preference data to fit the revealed preference data.

Before proceeding with the joint estimation one should consider if the two sets of data are derived from some similar underlying preference structure. We investigate this issue by plotting the predicted probabilities of choice, for each site, from the stated preference model against the same information from the revealed preference. Taking the logarithms of this information produces the graph in Fig. 2, in which the probabilities of site choice are estimated by residence zone. Clearly, there is a linear relationship between the log of the stated preference predicted

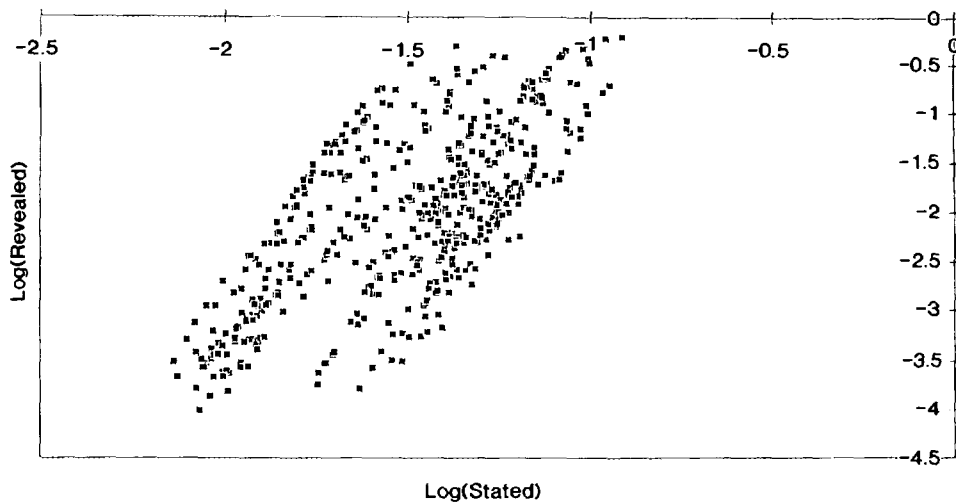


FIG. 2. Stated versus revealed preference predicted probabilities.

probabilities and the log of the revealed preference probabilities. Fitted lines between these points for each residence zone yield adjusted R^2 's between 0.297 and 0.588. Furthermore, a likelihood ratio test of the difference between parameters, incorporating the relative scale effect, yields a χ^2 (2 df) value of 0.003, implying that we cannot reject the hypothesis of equal parameters. Conditional on the assumption of equal parameters, a likelihood ratio test of equal scale factors yields a χ^2 (1 df) of 294.83. The null hypothesis of equal scale factors is rejected at a 99% level (see Swait and Louviere [34] for a discussion of these hypothesis tests). We consider this to be sufficient evidence that the stated and revealed preference data contain similar preference structures and that joint estimation, including the rescaling of one of the data series, is justified.

The data used in the revealed preference analysis and the stated preference analysis were recoded so that the same units were used for both data sets (i.e., effects coding rather than dummy variable coding was used for the discrete level attributes). The joint estimation results are in Table IV. The ρ^2 for this joint model was 0.68. All parameters had the same sign as the parameters in the stated preference model. Note that many more factors are significant in the joint model relative to the revealed preference model. This is probably because the colinearity in the revealed preference model has been reduced (or perfect colinearity no longer exists).

The welfare measures were recalculated using the parameters estimated from the joint model. These new measures are closer to the revealed preference model than the stated preference model. These results are presented in Table V.¹³

¹³The values in Table V are developed from unweighted estimation over all individuals in the sample. As described above, the respondents to the revealed preference component are the same individuals that responded to the stated preference component. In order to extend these results to provincial level aggregates, additional analysis was performed using weighted data and other forms of site attributes. These results, available in Alberta Environment [1], were based on weighted revealed preference results and unweighted stated preference results.

TABLE IV
Coefficients of the Joint Model

Attribute	Description	Standing water	Running water
Terrain ^a	Flat prairie versus	-1.557	-1.75
	Rolling prairie	-0.300	-0.422
		(0.194)	(0.179)
	Foothills	1.09	0.527
		(0.193)	(0.179)
	Mountain	0.767	1.65
		(0.191)	(0.179)
Fish size	Large (= 1) vs small (= -1)	0.244	0.380
		(0.111)	(0.104)
Fish catch	Fish per minute	0.262	0.428
		(0.113)	(0.106)
Fish species ^a	Mountain whitefish versus	n/a	-1.16
	Rainbow trout and mountain whitefish	n/a	0.271
			(0.181)
	Rainbow trout, mountain whitefish, and brown trout	n/a	0.453
			(0.176)
	Cutthroat trout, mountain whitefish, and bull trout	n/a	0.438
			(0.180)
Water quality	Good (= -1) vs bad (= 1)	-1.67	-1.36
		(0.113)	(0.104)
Facilities ^a	None versus	-1.51	-1.17
	Day-use only	-0.834	-0.465
		(0.194)	(0.184)
	Limited facility campsite	1.29	0.684
		(0.192)	(0.176)
	Fully serviced campsite	1.05	0.952
		(0.190)	(0.178)
Swimming	Yes (= -1) vs no (= 1)	-1.16	-0.670
		(0.112)	(0.104)
Beach	Yes (= -1) vs no (= 1)	-0.839	-0.522
		(0.112)	(0.104)
Distance	Kilometers	-0.029	-0.029
		(0.001)	(0.001)

^aAttributes with multiple levels are coded using effects codes. The base level is assigned -1 for all columns representing the remaining levels. Each column contains a 1 for the level represented by the column and a -1 for the base. The interpretation of these parameters is that the base level takes the utility level of the negative of the sum of the estimated coefficients and each other level takes the utility associated with the coefficient.

TABLE V
Welfare Measures from the Stated, Revealed, and Joint Models

Residence zone	Revealed preference	Stated preference	Joint model
1	0.77	4.43	0.40
2	1.55	5.67	1.16
3	1.04	4.60	0.59
4	1.24	5.18	0.67
5	0.64	4.62	0.38
6	1.59	4.63	0.68
7	1.68	5.16	0.86
8	0.46	4.29	0.21
9	0.58	4.44	0.32
10	0.66	6.33	0.33
11	0.69	4.66	0.41
12	0.53	4.64	0.35
13	2.83	6.79	2.74
14	1.48	5.20	0.85
15	1.56	5.79	1.06
16	1.20	6.55	2.02
17	1.78	7.73	2.60
18	1.97	7.08	3.28
19	2.80	7.43	4.15
20	3.99	7.96	5.29
21	2.46	8.06	2.72
22	2.00	7.05	2.04
23	1.07	7.03	1.96
24	1.05	7.27	0.96
25	0.65	5.95	0.56

DISCUSSION

The techniques used and findings presented in this paper should be of interest to economists for several reasons. First, as a method of valuing non-market goods, the choice experiment approach seems to be a very fruitful variant of stated preference techniques. The most common stated preference technique employed in economic analysis is contingent valuation. While contingent valuation has been shown to perform admirably in some settings, many other studies question its validity (see Cummings *et al.* [10] or Mitchell and Carson [29]). The choice experiment approach seems to be very flexible in terms of modeling complex trade-offs between attributes. In this particular study, we can also show that the underlying preferences reflected in the stated preference and revealed preference models are similar. This comparison is based on identical theories of choice, the random utility model. These findings lend support to the use of this stated preference technique, at least in the measurement of use values.

While others have promoted the use of stated preference techniques like conjoint analysis (Goodman [13], Mackenzie [23, 24]), we believe that this paper presents one of the first applications of joint stated and revealed preference methods in the non-market valuation literature. In related literature, Cameron [9] presents a model which jointly estimates contingent valuation and travel cost parameters. This approach differs from ours as we employ actual and stated

choices of recreation sites while Cameron combines actual recreational activity decisions with stated intentions to pay for a quality changes. Cameron's approach combines the information from the utility function used to analyze the actual choice decision with the utility function underlying the willingness to pay decision. In particular, the parameters of the utility function are estimated, based on information from both sets of data. In essence, we build upon the pioneering approach employed by Cameron by designing the stated and revealed preference questions to conform to a common random utility formation.¹⁴ In our joint model, all of the parameters are estimated using information from the stated and revealed preference information. In both our approach and Cameron's approach, however, an attempt is made to combine the information in the stated preference analysis and the revealed preference analysis.¹⁵

A second advance presented in this study is the use of the stated preference model to improve the quality of the estimates of the revealed preference model. Colinearity can be a significant problem in applications of revealed preference models, making it difficult to isolate the effects of attributes. In applied work it is often precisely this isolation of factors which is required. Furthermore, applications in non-market valuation often require simulation of conditions (attribute levels) outside the span of current conditions. Stated preference techniques can be used to analyze the response of individuals to attribute ranges not presently available.

Finally, contingent valuation has received considerable attention recently because of the controversy regarding non-use value estimates. While we present a case involving use values in this paper, an application to non-use values is relatively easy to construct. Non-use values have been measured using contingent valuation (Mitchell and Carson [29]) as well as contingent ranking techniques (Rae [31], Lareau and Rae [16], and Smith and Desvousges [33]). The appealing traits of the choice experiment technique (focused respondent tasks, an emphasis on trade-offs between factors, consistency with the random utility model) suggest that it may offer a valuable alternative to contingent valuation in the measurement of non-use values. For example, individuals could be presented with alternative community level attributes (e.g., air quality) as well as levels of household taxes paid. Such an analysis has been employed by Hensher [15] as a decision aid in planning transportation facilities. This appears to be a fruitful avenue for future research.

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¹⁴Cameron [9] suggests that valuation data sets "can (and should) be designed so that it is feasible to adopt the more rigorous 'random utility stochastic' specifications customarily used in the discrete/continuous choice literature" (p. 315).

¹⁵Dickie *et al.* [11] have also combined contingent valuation stated preference results and revealed preference results for a market good.

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